

Figure 1 is a plot showing the Dispersion Energy (kcal/mol) versus the number of basis functions (N) for the [2+2] cycloaddition of ethylene. The plot includes an inset showing a zoomed-in view of the region where N is between 3.0 and 3.5. The legend identifies three data series:

- SAPT(PBE0)-D4(S, ND) (Teal line with triangles)
- SAPT(PBE0)-D4(I, ND) (Orange line with triangles)
- SAPT2+3(CCD)/aTZ (Black line with squares)

The main plot shows that the dispersion energy increases with N , approaching a value near 0 kcal/mol. The inset highlights the rapid change in energy for small N values, where the SAPT(PBE0)-D4(S, ND) method shows a significant increase in energy as N increases from 3.0 to 3.5.

Figure 1 is a line plot showing the dispersion energy (in kcal/mol) as a function of distance (in Å) for the (H₂O)₈ cluster. The x-axis ranges from 3.0 to 6.5 Å, and the y-axis ranges from -15 to 0 kcal/mol. A vertical dashed line is drawn at 3.4 Å. The legend identifies the following methods:

- HF-D4(ATM) (blue circles)
- SAPT(PBE0)-D4(S) (teal diamonds)
- SAPT(PBE0)-D4(I) (orange diamonds)
- PBE0-D4/aDZ (red stars)
- SAPT0-D4(S) (purple circles)
- SAPT(PBE0)/aDZ (brown squares)
- SAPT(PBE0)/aTZ (grey squares)
- SAPT2+3(CCD)/aTZ (black squares)

The curves show that the dispersion energy increases (becomes less negative) as the distance increases, with all methods converging towards zero at larger distances. The SAPT(PBE0)-D4(S) method generally yields the most negative dispersion energy at shorter distances, while the HF-D4(ATM) method yields the least negative energy at shorter distances.